Mekanik.py Formula Sheet

A Python-Based Mechanical Engineering Formula Sheet

Analytical formulas and computational interfaces

Library name:	`mekanik.py`
Domain:	Mechanical engineering
Implementation:	Python
Version:	0.1

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Introduction

This document presents `mekanik.py`, a compact Python-based reference library for classical mechanical engineering calculations. Its purpose is to connect analytical formulas commonly found in engineering handbooks with their direct computational use in engineering analysis, design, and verification.

The content is organized as a formula sheet structured by topic. Each analytical expression is paired with a corresponding Python function, creating a clear one-to-one correspondence between theory and implementation. This structure emphasizes transparency and traceability, and all formulas are written using standard mechanical engineering notation and units.

The current version covers tolerances according to ISO 286, bolted joints, springs, interference fits, and basic models for brakes and wear. The focus is on clarity and correctness rather than numerical optimization or advanced numerical methods. As such, the library is well suited for preliminary design calculations, parametric studies, educational use, and the verification of numerical results.

The document is designed to be extensible. Additional chapters can be added using the same structure: a clear analytical formulation, a consistent set of symbols, and a minimal, explicit Python interface. In this way, `mekanik.py` serves both as a practical engineering tool and as a concise, self-contained mechanical engineering reference.

Contents

1 Tolerances

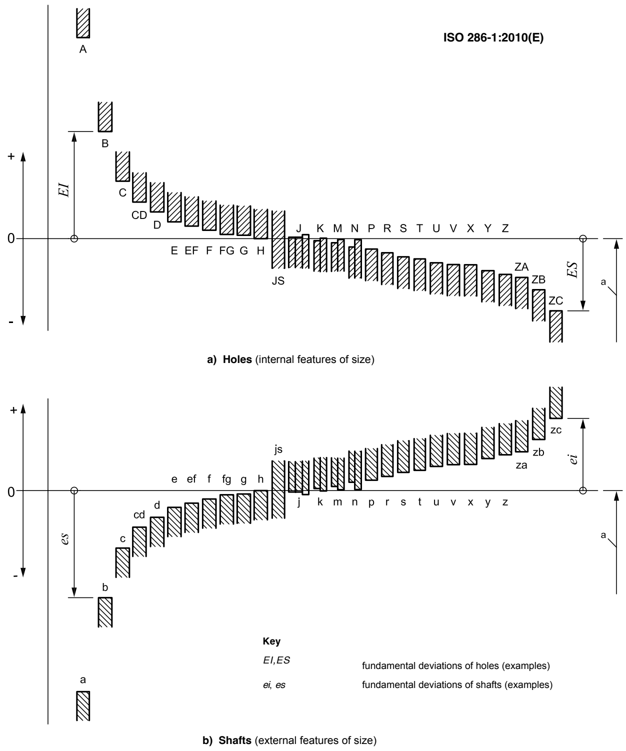


Figure 1: Schematic representation of tolerance positions

1.1 Tolerance definitions

$$\textbf{Tolerance (width): } T = U - L \tag{1}$$

$$\textbf{Deviations from nominal (hole): } ES = U - D, \qquad EI = L - D \tag{2}$$

$$\textbf{Deviations from nominal (shaft): } es = U - D, \qquad ei = L - D \tag{3}$$

1.2 IT grades

Note: Internally, IT values are handled in μm and converted to mm when combined with nominal dimensions.

$$\textbf{IT grade tolerance value: } IT = IT(D,\ g) \tag{4}$$

```
IT = it_um(nominal_mm, grade)   # IT in micrometers
```

1.3 Fundamental deviations

$$\textbf{Tolerance limits from designation (e.g. H7, h6, JS7): } (L,\ U) = \text{iso\_tol}(D,\ \text{tol}) \tag{5}$$

```
(lower, upper) = iso_tol(nominal_mm, tol, mode="abs")  # absolute limits [mm]
(low_dev, high_dev) = iso_tol(nominal_mm, tol, mode="dev")  # deviations [mm]

(lower, upper) = iso_tol(10, 'H7', "abs", "mm")
```

$$\textbf{Symmetric zone (JS/js):}\ \Delta_{\min} = -\frac{IT}{2}, \qquad \Delta_{\max} = +\frac{IT}{2} \tag{6}$$

$$\textbf{Generic construction from one deviation value + IT:}\ \begin{cases} L = D + \Delta_{\min} \\ U = D + \Delta_{\max} \end{cases} \tag{7}$$

1.4 Fits

$$\textbf{Fit clearance range (hole - shaft):}\ c_{\min} = L_h - U_s, \qquad c_{\max} = U_h - L_s \tag{8}$$

$$\textbf{Overlap of tolerance intervals (size space):}\ \ell_{\text{overlap}} = \max\Big(0,\ \min(U_h, U_s) - \max(L_h, L_s)\Big) \tag{9}$$

```
fit = fit_check(nominal_mm, hole_tol, shaft_tol)
# hole_tol example: "H7"
# shaft_tol example: "h6"
```

The returned dictionary from `fit_check` contains:

```
fit["hole_limits"]    # (Lh, Uh)
fit["shaft_limits"]   # (Ls, Us)
fit["clearance"]      # (c_min, c_max)
fit["overlap"]        # overlap length
fit["fit_type"]       # "clearance" / "transition" / "interference"
```

Table 1: Symbols used in ISO 286 tolerance calculations

Symbol	Description	Unit
D	Nominal size	mm
L	Lower limit size	mm
U	Upper limit size	mm
T	Tolerance width $(U - L)$	mm
ES	Upper deviation (hole) $(U - D)$	mm
EI	Lower deviation (hole) $(L - D)$	mm
es	Upper deviation (shaft) $(U - D)$	mm
ei	Lower deviation (shaft) $(L - D)$	mm
IT	IT grade tolerance value	μm or mm
FD	Fundamental deviation (ISO 286)	μm
L_h, U_h	Hole limits	mm
L_s, U_s	Shaft limits	mm
$c_{\min}$	Minimum clearance $(L_h - U_s)$	mm
$c_{\max}$	Maximum clearance $(U_h - L_s)$	mm
ℓ_{overlap}	Interval overlap length	mm

2 Bolts

2.1 Forces

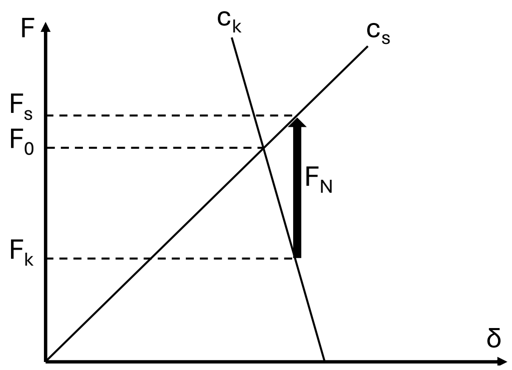


Figure 2: F-delta Diagram

$$\textbf{Screw force: } F_s = F_0 + \frac{c_s}{c_s + c_k} F_N \tag{10}$$

```
Fs = F_s(F0, FN, cs, ck)
```

$$\textbf{Clamping force: } F_k = F_0 - \frac{c_k}{c_s + c_k} F_N \tag{11}$$

```
Fk = F_k(F0, FN, cs, ck)
```

2.2 Torques

$$\textbf{Total tightening torque: } M = F_{\text{ax}} \left(0.16\, P + 0.58\, \mu_{\text{thread}}\, d_2 + \mu_{\text{head}}\, r_m\right) \tag{12}$$

```
M = M_total(Fax, P, d2, r_m, mu_head, mu_thread)
```

$$\textbf{Bearing friction torque: } M_b = \mu_b\, F_b\, r_m \tag{13}$$

```
M_b = M_bearing(mu_b, Fb, r_m)
```

$$\textbf{Thread tightening torque: } M_t = \frac{F_{\text{ax}}\, P\, \tan(\varphi + \rho')}{2\pi\, \tan\varphi} \tag{14}$$

```
M_t = M_tightening(Fax, r_ax, phi, rho, P)
```

$$\textbf{Thread loosening torque: } M_l = -\frac{F_{\text{ax}}\, P\, \tan(\varphi - \rho')}{2\pi\, \tan\varphi} \tag{15}$$

```
M_l = M_loosening(Fax, r_ax, phi, rho, P)
```

2.3 Geometry

$$\textbf{Apparent friction angle:}\ \rho' = \arctan\left(\frac{\mu}{\cos\alpha}\right) \tag{16}$$

```
rho = rho_apparent(mu, alpha)
```

$$\textbf{Lead angle:}\ \varphi = \arctan\left(\frac{P}{\pi d_2}\right) \tag{17}$$

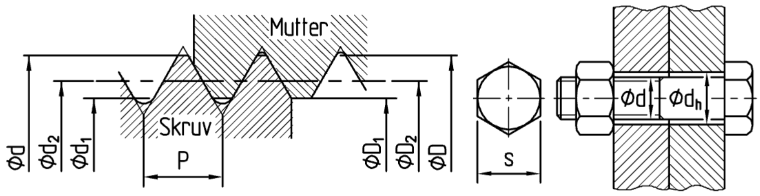
```
phi = phi_lead(P, d2)
```

$$\textbf{Thread stress area:}\ A_s \approx \frac{\pi}{16}\,(d_1 + d_2)^2 \tag{18}$$

```
As = A_s(d1, d2)
```

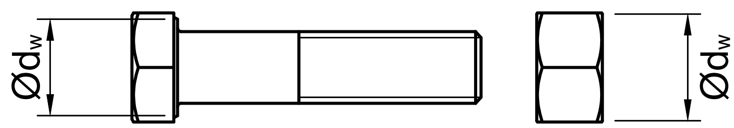
Metric thread dimensions: Lookup of standard ISO metric thread geometry from table **??** (19)

```
dims = M_dimensions(d_nom1)

P  = dims["P"]
d  = dims["d"]
d2 = dims["d2"]
d1 = dims["d1"]

dh_fine   = dims["dh_fine"]
dh_medium = dims["dh_medium"]
dh_coarse = dims["dh_coarse"]
```



Thread	P [mm]	d [mm]	d_2 [mm]	d_1 [mm]	$d_{h,\mathrm{f}}$ [mm]	$d_{h,\mathrm{m}}$ [mm]	$d_{h,\mathrm{c}}$ [mm]
M1	0.25	1	0.838	0.729	1.1	1.2	1.3
M1.2	0.25	1.2	1.038	0.929	1.3	1.4	1.5
M1.4	0.30	1.4	1.205	1.075	1.5	1.6	1.8
M1.6	0.35	1.6	1.373	1.221	1.7	1.8	2.0
M1.8	0.35	1.8	1.573	1.421	2.0	2.1	2.2
M2	0.40	2.0	1.740	1.567	2.2	2.4	2.6
M2.5	0.45	2.5	2.208	2.013	2.7	2.9	3.1
M3	0.50	3.0	2.675	2.459	3.2	3.4	3.6
M4	0.70	4.0	3.545	3.242	4.3	4.5	4.8
M5	0.80	5.0	4.480	4.134	5.3	5.5	5.8
M6	1.00	6.0	5.350	4.917	6.4	6.6	7.0
M8	1.25	8.0	7.188	6.647	8.4	9.0	10.0
M10	1.50	10	9.026	8.376	10.5	11.0	12.0
M12	1.75	12	10.863	10.106	13.0	13.5	14.5
M16	2.00	16	14.701	13.835	17.0	17.5	18.5
M20	2.50	20	18.376	17.294	21.0	22.0	24.0
M24	3.00	24	22.051	20.752	25.0	26.0	28.0
M30	3.50	30	27.727	26.211	31.0	33.0	35.0
M36	4.00	36	33.402	31.670	37.0	39.0	42.0
M42	4.50	42	39.077	37.129	43.0	45.0	48.0
M48	5.00	48	44.752	42.587	50.0	52.0	56.0
M56	5.50	56	52.428	50.046	58.0	62.0	66.0
M64	6.00	64	60.103	57.505	66.0	70.0	74.0

Figure 3: Metric thread dimensions. The schematic defines the geometric quantities used for ISO metric threads, and the table lists standard values for common thread sizes.



	M5	**M6**	**M8**	**M10**	**M12**	**M16**
d_w, bolt head [mm]	6.74	8.74	11.47	14.47	16.47	22.00
d_w, nut [mm]	6.90	8.90	11.47	14.60	16.60	22.50
	M20	**M24**	**M30**	**M36**	**M42**	**M48**
d_w, bolt head [mm]	27.70	33.25	42.75	51.11	59.95	69.45
d_w, nut [mm]	27.70	33.25	42.75	51.11	59.95	69.45

Figure 4: Effective contact diameter d_w for bolt head and nut. The schematic defines d_w, and the table lists standard values for common ISO metric threads.

Table 2: Symbols used in bolt calculations

Symbol	Description	Unit
F_s	Bolt force	N
F_k	Clamping force	N
F_0	Preload force	N
F_N	External axial load	N
F_{ax}	Axial bolt force	N
F_b	Bearing force	N
c_s	Bolt stiffness	N/m
c_k	Clamped parts stiffness	N/m
M	Total tightening torque	Nm
M_b	Bearing friction torque	Nm
M_t	Thread tightening torque	Nm
M_l	Thread loosening torque	Nm
P	Thread pitch	m
d_1	Minor diameter	m
d_2	Pitch diameter	m
r_m	Mean bearing / friction radius	m
φ	Thread lead angle	rad
ρ'	Apparent friction angle	rad
α	Thread flank angle	rad
μ	Thread friction coefficient	–
μ_{thread}	Thread friction coefficient	–
μ_{head}	Bearing (under-head) friction coefficient	–
μ_b	Bearing surface friction coefficient	–
A_s	Tensile stress area of thread	m^2

3 Springs

3.1 Spring stiffness

$$\textbf{Axial spring constant (helical spring): } c_a = \frac{G\,d^4}{8\,n\,D^3} \tag{20}$$

```
ca = spring_constant_axial(G, d, D, n)
```

$$\textbf{Torsional spring constant: } c_v = \frac{E\,d^4}{64\,n\,D} \tag{21}$$

```
cv = spring_constant_torsional(E, d, D, n)
```

3.2 Stresses

$$\textbf{Torsional shear stress: } \tau_v = \frac{8\,F\,D}{\pi\,d^3} \tag{22}$$

```
tau_v = shear_stress_torsion(F, D, d)
```

$$\textbf{Effective shear stress (with correction): } \tau_e = k\,\frac{8\,F\,D}{\pi\,d^3} \tag{23}$$

```
tau_e = shear_stress_effective(F, D, d, k)
```

$$\textbf{Wahl correction factor: } k = \frac{4c-1}{4c-4} + \frac{0.615}{c}, \qquad c = \frac{D}{d} \tag{24}$$

```
k = wahl_factor(D, d)
```

3.3 Geometry and stability

$$\textbf{Minimum free spring length: } l_0 = 1.25\,(n+1)\,d + \delta \tag{25}$$

```
l0 = free_length(n, d, delta)
```

$$\textbf{Buckling criterion for compression springs: } l_0 < 2.6\,D \tag{26}$$

```
stable = check_buckling(l0, D)
```

3.4 Energy and work

$$\textbf{External work (linear spring): } W = \frac{1}{2}\,F\,\delta \tag{27}$$

```
W = external_work_linear(F, delta)
```

$$\textbf{External work (rotational spring): } W = \frac{1}{2}\,M\,\varphi \tag{28}$$

```
W = external_work_rotational(M, phi)
```

$$\textbf{Maximum storable energy (normal stress):}\ W_{\max} = V\,\frac{\sigma_{\max}^2}{2\,E} \tag{29}$$

```
Wmax = max_energy_density_normal(sigma_max, E, V)
```

$$\textbf{Maximum storable energy (shear stress):}\ W_{\max} = V\,\frac{\tau_{\max}^2}{2\,G} \tag{30}$$

```
Wmax = max_energy_density_shear(tau_max, G, V)
```

$$\textbf{Utilization factor:}\ \eta = \frac{W_{\text{actual}}}{W_{\max}}, \qquad 0 < \eta < 1 \tag{31}$$

```
eta = utilization_factor(W_actual, W_max)
```

Table 3: Symbols for spring formulas

Symbol	Description	Unit
E	Young's modulus	Pa
G	Shear modulus	Pa
d	Wire diameter	m
D	Mean coil diameter	m
n	Number of active coils	–
c	Spring index, $c = D/d$	–
c_a	Axial spring constant (stiffness)	N/m
c_v	Torsional (rotational) spring constant	N·m/rad
F	Spring force (axial load)	N
M	Torque	N·m
δ	Axial deflection	m
φ	Angular deflection	rad
τ_v	Torsional shear stress (uncorrected)	Pa
τ_e	Effective shear stress (corrected)	Pa
k	Correction factor (Wahl factor)	–
l_0	Free spring length	m
W	External work / stored elastic energy	J
V	Loaded volume	m^3
$\sigma_{\max}$	Maximum normal stress	Pa
$\tau_{\max}$	Maximum shear stress	Pa
$W_{\max}$	Maximum storable elastic energy	J
W_{actual}	Actual stored elastic energy	J
η	Utilization factor	–

4 Interference fits

4.1 Friction forces

$$\textbf{Frictional force at macro-sliding: } F = \mu\,\pi\,d\,L\,p = \mu\,N \tag{32}$$

```
F = friction_force(mu, p=p, d=d, L=L)
# or
F = friction_force(mu, N=N)
```

$$\textbf{Resultant friction force: } F = \sqrt{F_{\text{ax}}^2 + F_{\text{per}}^2} \tag{33}$$

```
F = friction_force_resultant(F_axial, F_peripheral)
```

$$\textbf{Transmitted torque: } M = \frac{d}{2}\,F_{\text{per}} \tag{34}$$

```
M = torque_transmitted(F_peripheral, d)
```

4.2 Geometry relations

$$\textbf{Hub geometry ratio: } \kappa = \frac{d}{D} \tag{35}$$

```
k = kappa(d, D)
```

$$\textbf{Shaft geometry ratio (hollow shaft): } \kappa_0 = \frac{d_0}{d} \tag{36}$$

```
k0 = kappa0(d0, d)
```

4.3 Interference and contact pressure

$$\textbf{Interface pressure from interference: } p = \frac{\Delta}{d\left[\frac{1}{E_{\text{hub}}}\left(\frac{1+\kappa^2}{1-\kappa^2} + \nu_{hub}\right) + \frac{1}{E_{\text{shaft}}}\left(\frac{1+\kappa_0^2}{1-\kappa_0^2} + \nu_{shaft}\right)\right]} \tag{37}$$

```
p = interference_pressure(delta, d, d0, D, E_hub, E_shaft, nu_hub, nu_shaft)
```

$$\textbf{Required diametral interference: } \Delta = p\,d\left[\frac{1}{E_{\text{hub}}}\left(\frac{1+\kappa^2}{1-\kappa^2} + \nu_{hub}\right) + \frac{1}{E_{\text{shaft}}}\left(\frac{1+\kappa_0^2}{1-\kappa_0^2} + \nu_{shaft}\right)\right] \tag{38}$$

```
delta = required_interference(p, d, d0, D, E_hub, E_shaft, nu_hub, nu_shaft)
```

4.4 Stresses after press fit

$$\textbf{Maximum equivalent stress (hub, Tresca): } \sigma^{T}_{e,\max} = p\,\frac{2\kappa^2}{1-\kappa^2} \tag{39}$$

$$\textbf{Maximum equivalent stress (hub, von Mises): } \sigma^{vM}_{e,\max} = p\,\frac{4\kappa^2}{3(1-\kappa^2)}(1+\kappa^2) \tag{40}$$

```
sigma = stresses_after_fit(p, k, material="hub", criterion="Tresca"  # or "vonMises")
```

$$\textbf{Maximum equivalent stress (hollow shaft): } \sigma_{e,\max} = p\,\frac{2\kappa_0^2}{1-\kappa_0^2} \tag{41}$$

```
sigma = stresses_after_fit(p, k, k0, material="hollow_shaft")
```

$$\textbf{Maximum equivalent stress (solid shaft): } \sigma_{e,\max} = p \tag{42}$$

```
sigma = stresses_after_fit(p, k, material="solid_shaft")
```

4.5 Thermal assembly

$$\textbf{Required temperature difference for assembly: } \Delta T = \frac{\Delta}{\alpha\, d} \tag{43}$$

```
dT = required_temperature_difference(delta, d, alpha)
```

Table 4: Symbols for interference fit formulas

Symbol	**Description**	**Unit**
μ	Coefficient of friction	–
p	Interface (contact) pressure	Pa
d	Interface diameter / fit diameter	m
D	Hub outer diameter	m
d_0	Shaft inner diameter (hollow shaft)	m
L	Contact length	m
κ	Hub geometry ratio, $\kappa = d/D$	–
κ_0	Shaft geometry ratio, $\kappa_0 = d_0/d$	–
Δ	Diametral interference	m
F	Frictional (tangential) force	N
F_{ax}	Axial friction force component	N
F_{per}	Peripheral friction force component	N
N	Normal force at the interface, $N = \pi dLp$	N
M	Transmitted torque	N·m
E_{hub}	Young's modulus (hub)	Pa
E_{shaft}	Young's modulus (shaft)	Pa
ν_{hub}	Poisson's ratio (hub)	–
ν_{shaft}	Poisson's ratio (shaft)	–
$\sigma_{e,\max}$	Maximum equivalent (effective) stress after fit	Pa
$\sigma_{e,\max}^{T}$	Maximum equivalent stress (Tresca)	Pa
$\sigma_{e,\max}^{vM}$	Maximum equivalent stress (von Mises)	Pa
α	Linear thermal expansion coefficient	1/K
ΔT	Required temperature difference for assembly	K

5 Brakes

5.1 Wear

$$\textbf{Archard's wear law: } Q = \frac{K\,W\,L}{H} \tag{44}$$

```
Q = archard_wear_law(K, W, L, H)
```

where Q is the total volume of wear debris, K is a dimensionless wear coefficient, W is the normal load, L is the sliding distance, and H is the hardness of the softer contacting material.

5.2 Disc brakes

$$\textbf{Disc brake torque (new pads, uniform pressure): } M = \frac{2}{3}\,\mu\,F_E\,\frac{R_o^3 - R_i^3}{R_o^2 - R_i^2} \tag{45}$$

```
M = disc_brake_new_pads(mu, F_E, R_o, R_i)
```

$$\textbf{Disc brake torque (worn pads, uniform wear): } M = \mu\,F_W\,\frac{R_o + R_i}{2} \tag{46}$$

```
M = disc_brake_worn_pads(mu, F_W, R_o, R_i)
```

Table 5: Symbols for brake and wear formulas

Symbol	**Description**	**Unit**
Q	Total wear volume (wear debris produced)	m^3
K	Wear coefficient (dimensionless)	–
W	Normal load	N
L	Sliding distance	m
H	Hardness of the softer contacting material	Pa
μ	Coefficient of friction	–
M	Brake torque (per pad)	N·m
F_E	Normal force per pad (new pads, uniform pressure)	N
F_W	Normal force per pad (worn pads, uniform wear)	N
R_o	Outer radius of pad contact area	m
R_i	Inner radius of pad contact area	m